

AERIS-10 — 22 V PA drain supply, enable switch and per-PA pulse gating — DSN-PSU-01 Rev A — STATUS: PROPOSED DESIGN (block level)

Date 2026-10-09 · generated by tools/design_pa_supply_schematic.py · decision D-14 (00_DESIGN_BASIS.md) · sizing from tools/design_thermal.py
Block-level only: component-level schematic, layout and protection detail must be captured in KiCad (MDR-12). Part numbers are proposals (availability to be verified).

Sequencing (existing firmware + this module): VG −4 V (DAC5578) → EN/DIS_RFPA_VDD high → U3 ramps 22V_SW → TX_GATE pulses drain only during chirps. Power-down reverse. Thermal/RF reasoning: engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md.
GND: single star at the module; chassis bond at FL1 Y-caps. Module footprint 120 × 80 mm, 25 mm high, mounted on the rear wall (DSN-MECH-01).

